Proof

$$\nabla_{\xi_i} L(\dots) = C - (\mu_i + \alpha_i) = 0$$

$$\Rightarrow \alpha_i + \mu_i = C \quad \forall i$$

Proof

The final dual is

$$\max_{\alpha_1,\ldots,\alpha_N} \left\{ \sum \alpha_i - \frac{1}{2} \left(\sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \mathbf{x}^i . \mathbf{x}^j \right) \right\}$$

$$\text{s.t. } \sum_{i=1}^{N} \alpha_i y_i = 0$$

$$0 <= \alpha_i <= C, \quad i : 1 \ldots N$$

$(\alpha \odot y)^T K (\alpha \odot y)$

Gram matrix $X X^T$

$$\alpha \odot y = \begin{bmatrix} \alpha_1 y_1 \\ \vdots \\ \alpha_N y_N \end{bmatrix}$$

$u_i \geq 0$

$\alpha_i + u_i = C$

$K_{ij} = x_i^T x_j$

The above can be written as $\sum_{i=1}^{N} \alpha - \frac{1}{2} (\alpha \odot y)^T K (\alpha \cdot y)$ $F(\alpha)$

K is positive semi definite,

Hessian of $F(\alpha) = -K \leftarrow$ negative semi-definite

$\Rightarrow F(\alpha)$ is concave in α.

Challenges in Solving the Dual

- Quadratic Programming (QP) with $\cancel{n}\,N$ variables
- Equality constraint: $\sum_i \alpha_i y_i = 0$
- Box constraints: $0 \leq \alpha_i \leq C$

Naive QP solvers:

- Require large memory ($O(\cancel{n}\,N^2)$ kernel matrix) ; Time required is between $O(N^2)$ & $O(N^3)$
- Scale poorly to large datasets

Solution: Sequential Minimal Optimization (SMO)

Core Idea of SMO

Observation: The equality constraint couples the α_i's.

Key Idea:

- Optimize *two* Lagrange multipliers at a time
- Enforce equality constraint analytically
- Reduce QP to a ~~2~~1-variable problem

One
~~Two~~-variable QP has a **closed-form solution**.

Details in this paper: John Platt. *Sequential Minimal Optimization: A Fast Algorithm for Training Support Vector Machines*. Microsoft Research, 1998.

Kernel Trick: extending to non-linear kernels

In the dual objective replace $\mathbf{x}^i.\mathbf{x}^j$ with $K(\mathbf{x}^i, \mathbf{x}^j)$.

Same trick in $f(\mathbf{x}) = \mathbf{w}.\mathbf{x} + w_0$ where $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}^i$ ...

$$\max_{\alpha_i} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \underbrace{x^i . x^j}$$

replace this term is kernel.

$$\Leftrightarrow \max_{\alpha_i} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j}^{N} \alpha_i \alpha_j y_i y_j k(x^i, x^j)$$

During train.

For inference:

$$f(x) = \underline{w} \cdot x + w_0 = \sum_{i=1}^{N} \alpha_i y_i x^i . x + w_0 = \sum_{i=1}^{N} \alpha_i y_i k(x^i, x) + w_0$$

replace with kernel

Demo

https://drive.google.com/file/d/1F7dLwQz4dMSf6UI0W3HW93IzIFsDr10t/view?usp=sharing

RBF kernel $k(x, x') = e^{\underline{-\gamma \|x - x'\|^2}}$

Why does Hinge Loss give rise to Sparse Solutions?

We will apply KKT conditions to show that all training examples for which $y_i f(\mathbf{x}^i) \geq 1$ will have corresponding $\alpha_i = 0$.

① $\nabla_w L(\cdot\cdot) = 0$

a) $w^* = \sum_{i=1}^{N} \alpha_i^* y_i x^i$

b) $\sum_{i=1}^{N} \alpha_i y_i = 0$

c) $C = \mu_i + \alpha_i$

② Complementary slackness

$u_i \cdot g_i(w) = 0$

a) $\alpha_i^* (1 - \xi_i^* - y_i(w^T x^i + w_0)) = 0$

b) $\mu_i^* \xi_i^* = 0$

Case 1 Consider examples where

$y_i(w^T x^i + w_0) > 1 \Rightarrow \xi_i^* = 0$

From 2(a) $\Rightarrow \alpha_i^* = 0$

Support vectors are those where $\alpha_i^* \neq 0$ and these can only be from one of the remaining 2 cases

Case 2

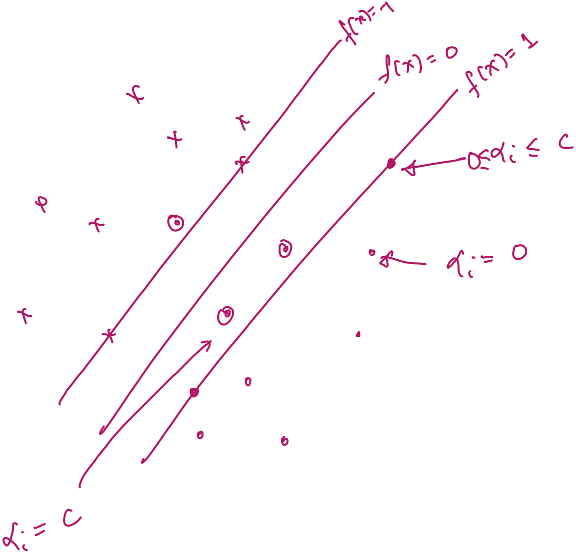
$y_i(w^T x^i + w_0) < 1 \Rightarrow \xi_i^* > 0$

From 2(b) $\mu_i^* = 0$

From 1(c) $\Rightarrow \alpha_i^* = C$

Case 3 $y_i(w^T x^i + w_0) = 1 \Rightarrow \xi_i^* = 0$

$\Rightarrow 0 \leq \alpha_i^* \leq C$

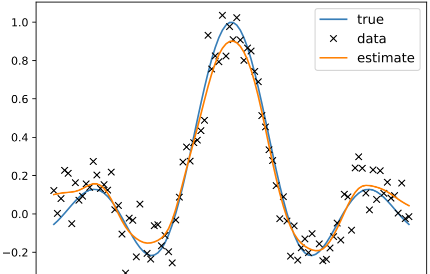

$f(x) = -1$
$f(x) = 0$
$f(x) = 1$
$0 \leq \alpha_i \leq C$
$\alpha_i = 0$
$\alpha_i = C$

Kernel Regression

Predict real-values based on similarity weighted values of training examples.

Nadaraya-Watson Kernel regression: (Equivalent to nearest neighbor classifiers for regression)

$$f(\mathbf{x}|D) = \sum_{i=1}^{N} \frac{K(\mathbf{x},\mathbf{x}^i)}{\sum_{j=1}^{N} K(\mathbf{x},\mathbf{x}_j)} y_i$$



Kernel ridge regression

General formulation of the regression problem

$f(x) = w^T x + w_0$

$\hat{y}_n = w^T x_n + w_0$

$x_n \in \mathbb{R}^D$

$$\longrightarrow F(w, \lambda) = \lambda \|w\|^2 + \sum_{n=1}^{N} \ell(y_n, \hat{y}_n)$$

We studied the L2 loss.

It can be shown that for this case the ridge regression problem can also be expressed in Kernel form:

continuous regression labels.

Recall that the closed form solution is:

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_D)^{-1} \mathbf{X}^\top \mathbf{y}$$

$X_{N \times D}$

$y_{N \times 1} \in \mathbb{R}^N$

Using the Matrix Inversion Lemma:

$XX^T \in \mathbb{R}^{N \times N}$

$(XX^T)_{ij} = \underbrace{x^i \cdot x^j}_{\to\, K(x^i, x^j)}$

$XX^T \equiv K$

$$= \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \lambda \mathbf{I}_N)^{-1}\mathbf{y}$$

$$= \mathbf{X}^\top(\mathbf{K} + \lambda \mathbf{I}_N)^{-1}_{N\times N}\mathbf{y}_{N\times 1}$$

Let $(K + \lambda I_N)^{-1} Y_{N\times 1} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{bmatrix} = \alpha$

$$= \underline{\mathbf{X}^\top \boldsymbol{\alpha}_{N\times 1}}$$

$$w^* = \sum_{i=1}^{N} \alpha_i \mathbf{x}_i$$

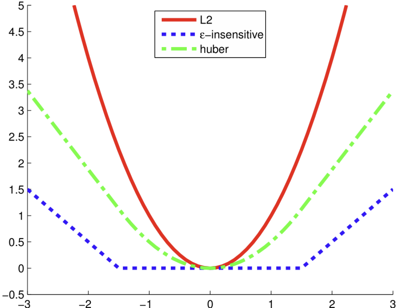
$f(x) = w^* \cdot x + w_0$

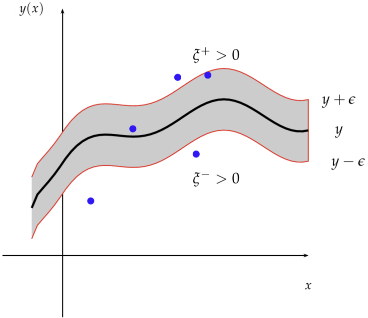
$= \sum_{i=1}^{N} \alpha_i \underbrace{x^i \cdot x}_{\text{kernel trick}} + w_0$

$= \sum_{i=1}^{N} \alpha_i K(x^i, x) + w_0$

Sparse Loss Functions

$$L_\epsilon(y, \hat{y}) \triangleq \begin{cases} 0 & \text{if } |y - \hat{y}| < \epsilon \\ |y - \hat{y}| - \epsilon & \text{otherwise} \end{cases}$$





SVM-like objective

(A)

$$J = \frac{1}{2}\|w\|^2 + C\sum_{n=1}^{N}\left(\xi_n^+ + \xi_n^-\right)$$

Show that (A) is equivalent to n

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}\max(0, |y-\tilde{y}|-\epsilon)$$

such that

$$\tilde{y}_n \leq f(x_n) + \epsilon + \xi_n^+$$

$$\tilde{y}_n \geq f(x_n) - \epsilon - \xi_n^-$$

Summary

- Kernel methods provide a new way to represent data and build models based purely on comparision of two points.
- Various kernels are available, each with its own characteristics.
- Kernel methods find applications in several ML algorithms: classifiers (Nearest neighbor, few-shot classification, SVMs) and regression (Kernel regression, support vector regression, Gaussian processes)